

Noorwali Khan

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ABOUT ME

Computer Science graduate from FUUAST Islamabad, with a CGPA of 3.75/4.00 and hands-on experience developing AI and computer vision systems. Developed VisionSlide, an offline gesture and voice-controlled presentation system, and an AI-Based Real-Time Video Analytics System for object detection, tracking, counting and spatial-temporal video analysis.

RESEARCH INTERESTS

Machine Learning | Deep Learning | Computer Vision | Image & Video Analysis | 3D Vision | Multimodal AI | Intelligent Systems

EDUCATION & TRAINING

BS Computer Science

FUUAST Islamabad | CGPA 3.75/4.00 | 2026

PROJECTS

VisionSlide: AI-Enabled Gesture and Voice Controlled Presentation System | Final Year Project

Developed a fully offline desktop application for touch-free presentation control using real-time hand gestures and offline speech recognition.

Technologies: Python | PySide6 | OpenCV | MediaPipe | Vosk | SQLite | PyInstaller

Key Contributions:

- Implemented real-time hand gesture and offline speech recognition.
- Developed a PySide6 GUI for presentation control and system feedback.
- Integrated automated slide navigation through system-level input simulation.
- Combined vision and speech-based interaction into a multimodal system.

Research Focus: Computer Vision | Multimodal AI | Human-Computer Interaction

Link: <https://github.com/Noor72353/VisionSlide.git>

AI-Based Real-Time Video Analytics System

Developed a desktop application for real-time and recorded video analysis using object detection, multi-object tracking, and spatial-temporal analytics.

Technologies: Python | PySide6 | OpenCV | Ultralytics YOLO | ByteTrack

Key Contributions:

- Implemented YOLO-based object detection and ByteTrack multi-object tracking.
- Developed object counting, line-crossing, zone, dwell-time, and trajectory analytics.
- Built a PySide6 interface for camera and video-based analysis.
- Packaged the system as a standalone Windows application with an installer.

Research Focus: Computer Vision | Object Detection | Multi-Object Tracking | Video Analytics

Link: <https://github.com/Noor72353/ai-video-analytics.git>

SKILLS

Programming Languages

Python | Java | C++

Artificial Intelligence & Machine Learning

Machine Learning | Deep Learning | Computer Vision | Image Processing | Data Science | Data Mining

Computer Vision & Video Analytics

OpenCV | Ultralytics YOLO | MediaPipe | ByteTrack | Object Detection & Tracking | Video Analytics

Speech & Multimodal Interaction

Vosk Speech Recognition | Gesture Recognition | Speech-Based Interaction | Multimodal Interaction

Frameworks & Libraries

PySide6 | NumPy

Development Tools

Git | GitHub | Visual Studio Code | PyInstaller

Databases

SQLite | Database Management Systems

HONOURS AND AWARDS

Prime Minister’s Youth Laptop Award | HEC Pakistan | 2023

Academic Excellence Award | FUUAST Islamabad | 2023

Merit-Based Partial Scholarship | FUUAST Islamabad | 2024

CERTIFICATIONS

Python for Everybody Specialization | University of Michigan, Coursera | 2025

Link: <https://coursera.org/verify/specialization/UUXT0E7ESRXZ>

LANGUAGE SKILLS

Mother tongue(s): Pashto | Urdu

Other language(s): English (Professional Working Proficiency)